

主机内核参数优化

1 主机内核参数优化

压测过程中为了提高服务器连接数的限制和并发性能，压力机客户端和服务端均可以按照如下的配置进行内核参数的调整

1.1 修改limits.conf文件

编辑/etc/security/limits.conf 文件，新增或修改为以下配置

代码块

```
1 * soft core unlimited
2 * hard core unlimited
3 * soft nofile 1048576
4 * hard nofile 1048576
5 * soft nproc 655360
6 * hard nproc 655360
7 * soft sigpending 255983
8 * hard sigpending 255983
```

1.2 修改/etc/sysctl.conf 文件

编辑/etc/sysctl.conf 文件，新增或修改为以下配置

代码块

```
1 # kernel
2 kernel.unknown_nmi_panic = 0
3 kernel.randomize_va_space = 0
4 kernel.msgmnb = 65536
5 kernel.msgmax = 65536
6 kernel.shmmax = 68719476736
7 kernel.shmall = 4294967296
8 kernel.panic = 60
9 kernel.panic_on_oops = 0
10 kernel.core_pattern = /opt/crash/core-%e-%s-%u-%g-%p-%t
11
12 # vm
13 vm.panic_on_oom = 0
14 vm.swappiness = 0
15 vm.zone_reclaim_mode = 0
16
```

```
17 # fs
18 fs.aio-max-nr = 1048576
19 fs.file-max = 1048576
20 fs.inotify.max_user_watches = 10000000
21 fs.inotify.max_queued_events = 327679
22
23 # net core
24 net.core.rmem_default = 8388608
25 net.core.wmem_default = 8388608
26 net.core.rmem_max = 16777216
27 net.core.wmem_max = 16777216
28
29 # net ipv4
30 net.ipv4.neigh.default.gc_stale_time=120
31 net.ipv4.ip_forward =0
32 net.ipv4.conf.all.rp_filter=0
33 net.ipv4.conf.default.rp_filter=0
34 net.ipv4.conf.default.arp_announce = 2
35 net.ipv4.conf.lo.arp_announce=2
36 net.ipv4.conf.all.arp_announce=2
37 net.ipv4.conf.all.send_redirects = 0
38 net.ipv4.conf.default.send_redirects = 0
39 net.ipv4.conf.all.secure_redirects = 0
40 net.ipv4.conf.default.secure_redirects = 0
41 net.ipv4.conf.all.accept_redirects = 0
42 net.ipv4.conf.default.accept_redirects = 0
43 net.ipv4.neigh.default.gc_thresh1 = 2048
44 net.ipv4.neigh.default.gc_thresh2 = 4096
45 net.ipv4.neigh.default.gc_thresh3 = 8192
46 net.ipv4.tcp_mem = 1000000 1500000 2000000
47 net.ipv4.tcp_rmem = 4096 16384 16777216
48 net.ipv4.tcp_wmem = 4096 16384 16777216
49 net.ipv4.tcp_max_syn_backlog = 8192
50 net.ipv4.tcp_fin_timeout = 10
51 net.ipv4.tcp_abort_on_overflow = 1
52 net.ipv4.tcp_syn_retries = 2
53 net.ipv4.tcp_synack_retries = 2
54 net.ipv4.tcp_retries2 = 5
55 net.ipv4.tcp_orphan_retries = 3
56 net.ipv4.tcp_syncookies = 1
57 net.ipv4.tcp_timestamps = 1
58 net.ipv4.tcp_tw_recycle = 0
59 net.ipv4.tcp_tw_reuse = 1
60 net.ipv4.tcp_max_tw_buckets = 200000
61 net.ipv4.ip_local_port_range = 5000 65535
62 net.ipv4.tcp_keepalive_intvl = 30
63 net.ipv4.tcp_keepalive_probes = 3
```

```
64 net.ipv4.tcp_keepalive_time = 1200
65 kernel.sysrq=1
66 net.core.somaxconn = 4096
67 net.netfilter.nf_conntrack_max = 500000
68 net.core.optmem_max = 16777216
69 net.core.netdev_max_backlog = 16384
70 net.core.netdev_max_backlog = 16384
71 # END
```